R Shiny

Benoît Thieurmel and Laurent Rouvière

2022-10-09

Contents

0.1 Summary . . . 3

1 Introduction **4**
1.1 Shiny: defining web applications with R . . . 4
1.2 My first application . . . 4

2 Starting with Rstudio **5**
2.1 One folder with one file . . . 5
2.2 One folder with one file . . . 6
2.3 One folder with two files . . . 6
2.4 One folder with two files . . . 7
2.5 Data/additional files . . . 7

3 Interactivity and communication **7**
3.1 Introduction | Example . . . 7
3.2 Introduction | process . . . 8
3.3 Notice . . . 9
3.4 UI part (input definition) . . . 9
3.5 Server part (output construction) . . . 9
3.6 UI part (output visualization) . . . 10
3.7 Back on the process . . . 10
3.8 Sharing ui <-> server . . . 10

4 Inputs **11**
4.1 Global view . . . 11
4.2 Numeric . . . 11
4.3 Characters . . . 11
4.4 Unique choice in a list . . . 12
4.5 Multiple choices in a list . . . 12
4.6 Simple Checkbox . . . 13
4.7 Multiple checkboxes . . . 13
4.8 Radio buttons . . . 13
4.9 Date | *Code* . . . 14
4.10 Date | *App* . . . 14
4.11 Period | *Code* . . . 14
4.12 Period | *App* . . . 15
4.13 Numeric slider numérique: one value . . . 15
4.14 Numeric slider: range . . . 15
4.15 Importing a file . . . 15
4.16 Action button . . . 16
4.17 Taking things further: building an input . . . 16

5 Outputs **17**
5.1 Global view . . . 17
5.2 Rules to define outputs . . . 17
5.3 Print . . . 17
5.4 Text . . . 17
5.5 Plot | *Code* . . . 17
5.6 Plot | *App* . . . 18
5.7 Table | *Code* . . . 18
5.8 Table | *App* . . . 19
5.9 DataTable | *Code* . . . 19
5.10 DataTable | *App* . . . 19
5.11 Defining UI elements in the SEVER | Process . . . 19
5.12 Defining UI elements in the SEVER | A simple example . . . 19
5.13 Taking things further: building an output . . . 20

6 Organizing the application **20**
6.1 **sidebarLayout** | Definition . . . 20
6.2 **sidebarLayout** | Example . . . 21
6.3 **navbarPage** | Definition . . . 21
6.4 **navbarPage** | App . . . 21
6.5 **navbarPage** | with `navbarMenu` . . . 21
6.6 **navbarPage** | Shiny app . . . 22
6.7 **tabsetPanel** | Definition . . . 22
6.8 **tabsetPanel** | Example . . . 22
6.9 **navlistPanel** | Definition . . . 23
6.10 **navlistPanel** | Example . . . 23
6.11 **Grid Layout** | Definition . . . 23
6.12 **Grid Layout** | Shiny app . . . 23
6.13 **wellPanel** | Definition . . . 23
6.14 **wellPanel** | Example . . . 24
6.15 **Combine structures** | Shiny app . . . 24
6.16 **shinydashboard** . . . 24

7 Interactive charts **25**
7.1 Introduction . . . 25
7.2 Integration in shiny . . . 25
7.3 Examples for server and ui functions . . . 25
7.4 Interactive charts: example . . . 26
7.5 Interactive charts: example . . . 26
7.6 Interactive charts: example . . . 27
7.7 Interactive charts: example . . . 27

8 Isolation **27**
8.1 Definition . . . 27
8.2 Example | ui.R . . . 27
8.3 Example | server.R . . . 28
8.4 Example | App . . . 28

9 Reactive expressions **29**
9.1 Definition . . . 29
9.2 Without reactive expressions . . . 29
9.3 With a reactive expression . . . 29
9.4 Notes . . . 29

10 Observe & functions to update **30**

10.1 Observe & fonctions to update . . . 30
10.2 Example for an input | App . . . 30
10.3 Example for tabs | App . . . 31
10.4 Example for an input | ui.R . . . 31
10.5 Example for an input | server.R . . . 32
10.6 Example for an input | App . . . 32
10.7 Example for tabs | ui.R . . . 33
10.8 Example for tabs | server.R . . . 33
10.9 Example for tabs | App . . . 34
10.10 ObserveEvent . . . 34

11 Conditional panels **34**
11.1 Definition . . . 34
11.2 Example for an input . . . 35
11.3 Example for an input . . . 35

12 Taking things further: HTML / CSS **35**
12.1 **Including HTML** . . . 35
12.2 **Some interested tags** . . . 37
12.3 CSS: introduction . . . 37
12.4 HTML / CSS | css external file . . . 37
12.5 HTML / CSS | css in the header . . . 38
12.6 HTML / CSS | CSS in an element . . . 39

13 Taking things further: some important "rules" **39**
13.1 Good approach . . . 39
13.2 Good approach . . . 40

14 Taking things further: debugging **40**
14.1 Printing in the console . . . 40
14.2 Printing in the console . . . 40
14.3 Manual launch of a browser . . . 41
14.4 Manual launch of a browser . . . 41
14.5 Automatic launch of a browser . . . 41
14.6 Mode "showcase" . . . 41
14.7 Mode "showcase" . . . 41
14.8 Reactive log . . . 42
14.9 Reactive log . . . 42
14.10 Communication in the server . . . 42
14.11 Communication in the server . . . 43
14.12 Error tracking . . . 43
14.13 Error tracking . . . 43

15 References **43**
15.1 Tutorials / Examples . . . 43

0.1 Summary

1. Introduction
2. Starting with Rstudio
3. Interactivity and communication
4. Inputs & outputs
5. Organizing the page
6. Interactive charts
7. HTML / CSS

8. More

1 Introduction

1.1 Shiny: defining web applications with R

Shiny is a **R** package that makes it easy to build interactive web applications with **R**

- does not require web expertise
- combine datascience power of **R** with web interactivity
- create local applications
- or deploy applications for other users: **shiny-server**, **shinyapps.io**, **shinyproxy**

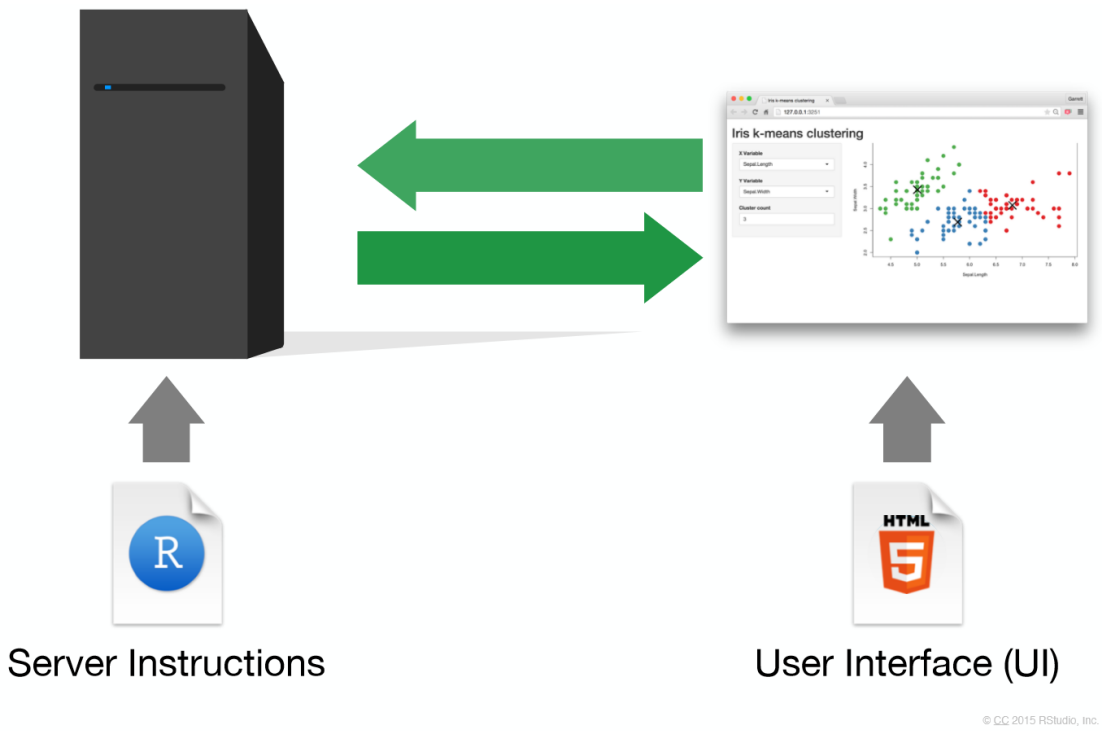
http://shiny.rstudio.com

http://www.shinyapps.io/

https://www.shinyproxy.io/

https://www.rstudio.com/products/shiny/shiny-server/.

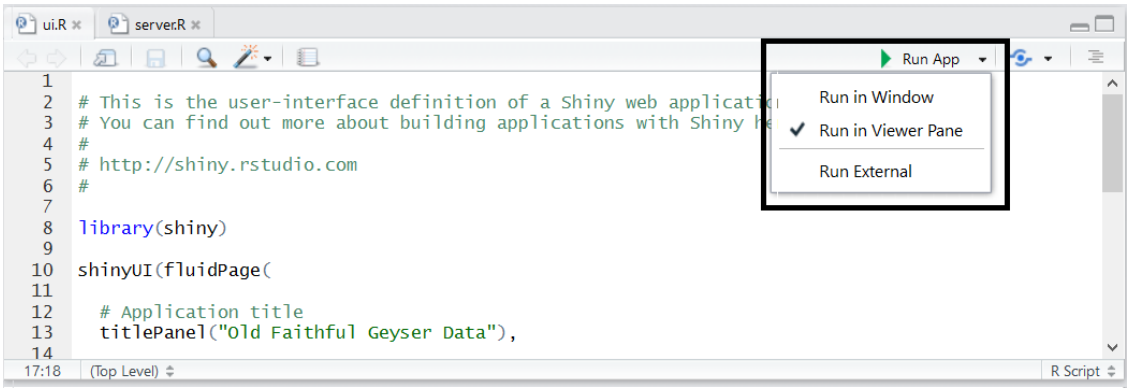
A shiny web applications requires a computer/server with **R**



1.2 My first application

- Starting an application is easy with **RStudio**, just start a **new project**
  - File -> New Project -> New Directory -> Shiny Web Application
  - Or File -> New File -> Shiny Web App -> Multiple File
  - Based on two scripts: **ui.R** and **server.R**
- Useful commands:
  - run the application: button **Run app**
  - update: button **Reload app**

– stop: button **Stop**



- **Run in Window**: new window, using **RStudio** environment
- **Run in Viewer Pane**: tab *Viewer* of **RStudio**
- **Run External**: in the default web browser

---

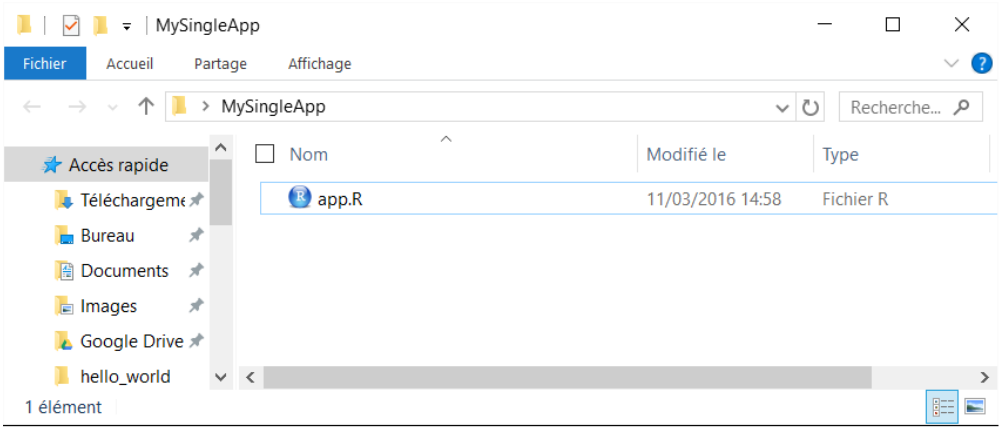
2 Starting with Rstudio

2.1 One folder with one file

Conventions:

- save as **app.R**

- end with shinyApp() command
- for **small applications**



2.2 One folder with one file

```
library(shiny)
ui <- fluidPage(
    titlePanel("Old Faithful Geyser Data"),
    sidebarLayout(
        sidebarPanel(
            sliderInput("bins","Number of bins:",
                        min = 1,max = 50,value = 30)
        ),
        mainPanel(
           plotOutput("distPlot")
        )
    )
)

server <- function(input, output) {

    output$distPlot <- renderPlot({
        x    <- faithful[, 2]
        bins <- seq(min(x), max(x), length.out = input$bins + 1)
        hist(x, breaks = bins, col = 'darkgray', border = 'white',
             xlab = 'Waiting time to next eruption (in mins)',
             main = 'Histogram of waiting times')
    })
}


shinyApp(ui = ui, server = server)
```

2.3 One folder with two files

Conventions:

- user interface (layout and appearance) in **ui.R**
- **R** instructions needed to build the app in **server.R**
- best structure for **complex applications**

2.4 One folder with two files

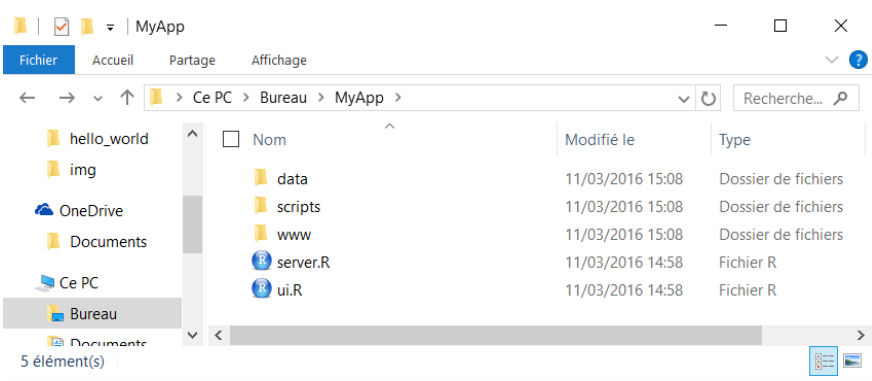
ui.R

```
library(shiny)
shinyUI(fluidPage(
    titlePanel("Old Faithful Geyser Data"),
    sidebarLayout(
        sidebarPanel(
            sliderInput("bins","Number of bins:",
                        min = 1,max = 50,value = 30)),
        mainPanel(plotOutput("distPlot")))
))
```

server.R

```
library(shiny)
shinyServer(function(input, output) {
    output$distPlot <- renderPlot({
        x    <- faithful[, 2]
        bins <- seq(min(x), max(x), length.out = input$bins + 1)
        hist(x, breaks = bins, col = 'darkgray', border = 'white',
             xlab = 'Waiting time to next eruption (in mins)',
             main = 'Histogram of waiting times')})
})
```

2.5 Data/additional files

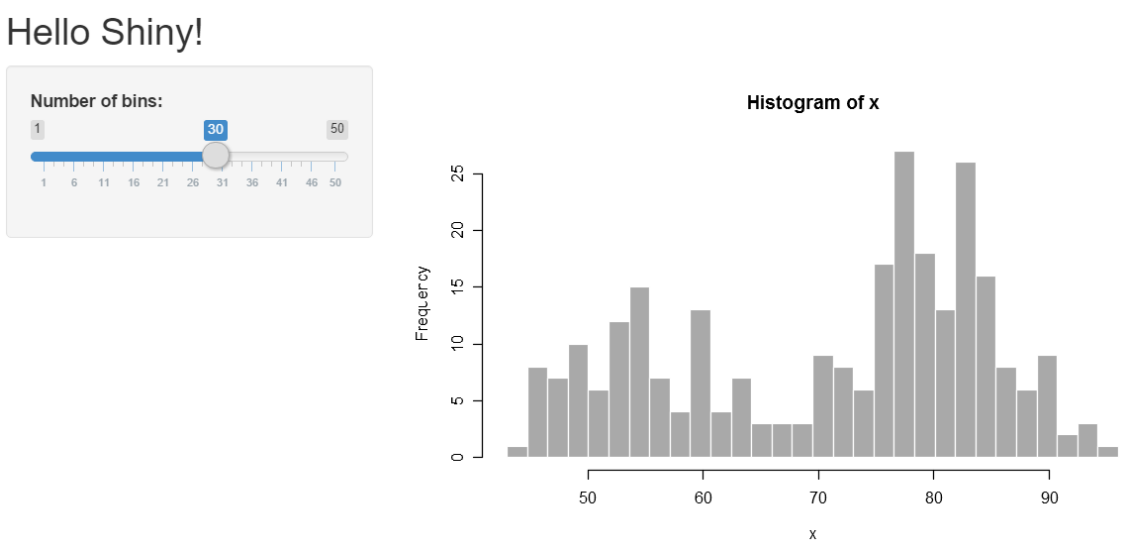


3 Interactivity and communication

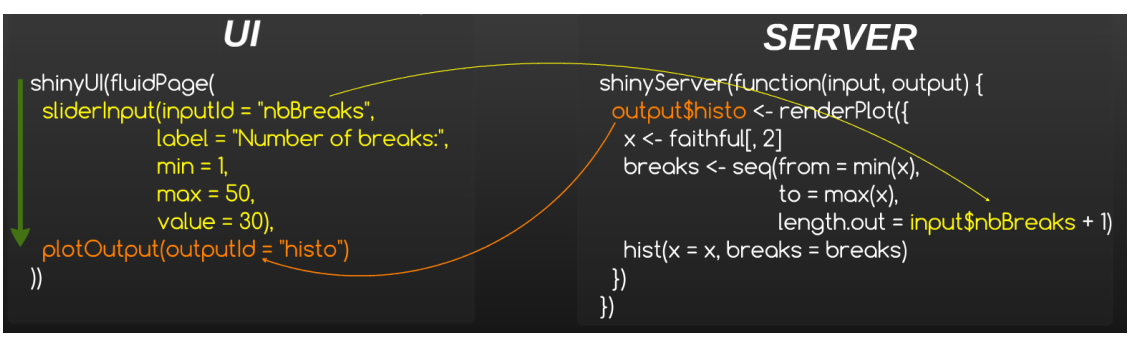
3.1 Introduction | Example

```
shinyApp(
  ui = fluidPage(
    titlePanel("Hello Shiny!"),
    sidebarLayout(
      sidebarPanel(
        sliderInput("bins",
                    "Number of bins:",
                    min = 1,
                    max = 50,
                    value = 30)
```

```
    ),
    mainPanel(
      plotOutput("distPlot")
    )
  )
),

server = function(input, output) {

  # Expression that generates a histogram. The expression is
  # wrapped in a call to renderPlot to indicate that:
  #
  #  1) It is "reactive" and therefore should be automatically
  #     re-executed when inputs change
  #  2) Its output type is a plot

  output$distPlot <- renderPlot({
    x    <- faithful[, 2]  # Old Faithful Geyser data
    bins <- seq(min(x), max(x), length.out = input$bins + 1)
    # draw the histogram with the specified number of bins
    hist(x, breaks = bins, col = 'darkgray', border = 'white')
  })
})
```



3.2 Introduction | process



- **ui**: organize inputs and outputs
- **server**: compute the outputs (from the inputs)
- **Server and ui communicate through inputs and outputs**
- **By default an output is updated as soon as an input changes**

3.3 Notice

Definition of the user interface: UI

- definition of the inputs
- architecture of the page, with location of the outputs

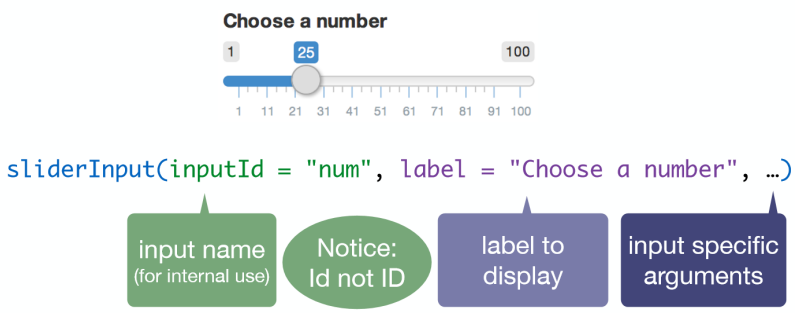
server/computing part: SERVER

- definition and computation of the outputs

3.4 UI part (input definition)

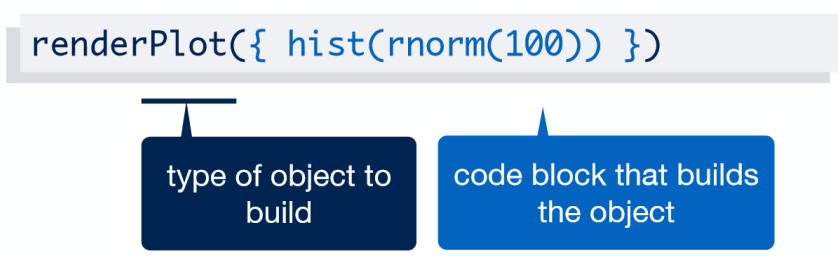
Two kinds of items in UI

- xxInput(inputId = . . . , . . .):
  - for an element which requires an action of the user
  - available in the server through its ID **input$inputId**



3.5 Server part (output construction)

- renderXX({expr}):
  - compute and return an output (which can depend on inputs) with classical **R** commands

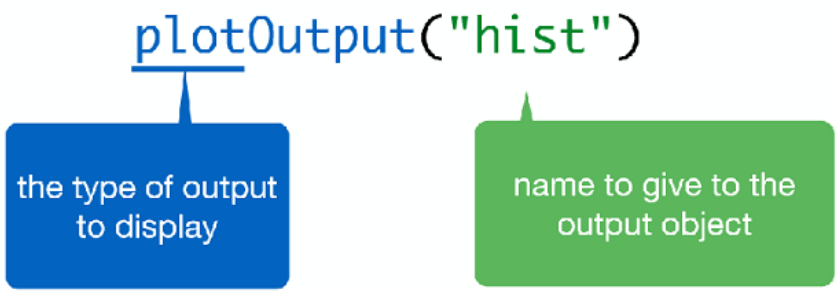


- Example:

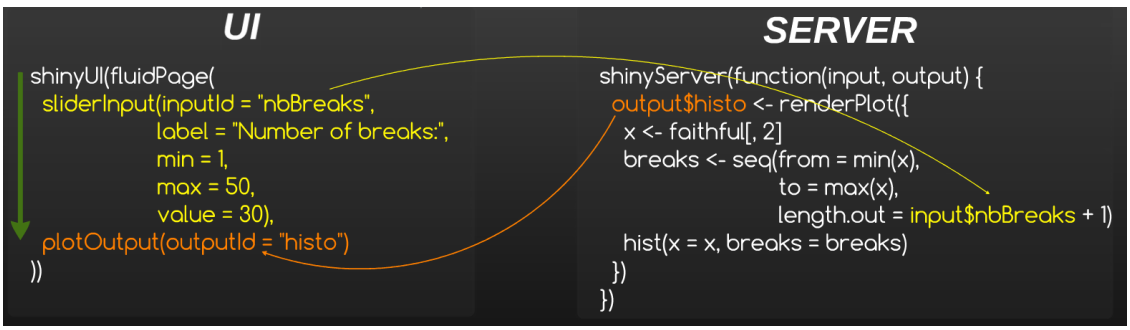
```
output$hist <- renderPlot({
  #commands to build the histogram
})
```

3.6 UI part (output visualization)

- xxOutput(ouputId = . . .):
  - refer to an output created in the server
  - often for graphs and/or tables



3.7 Back on the process



Is it clearer?

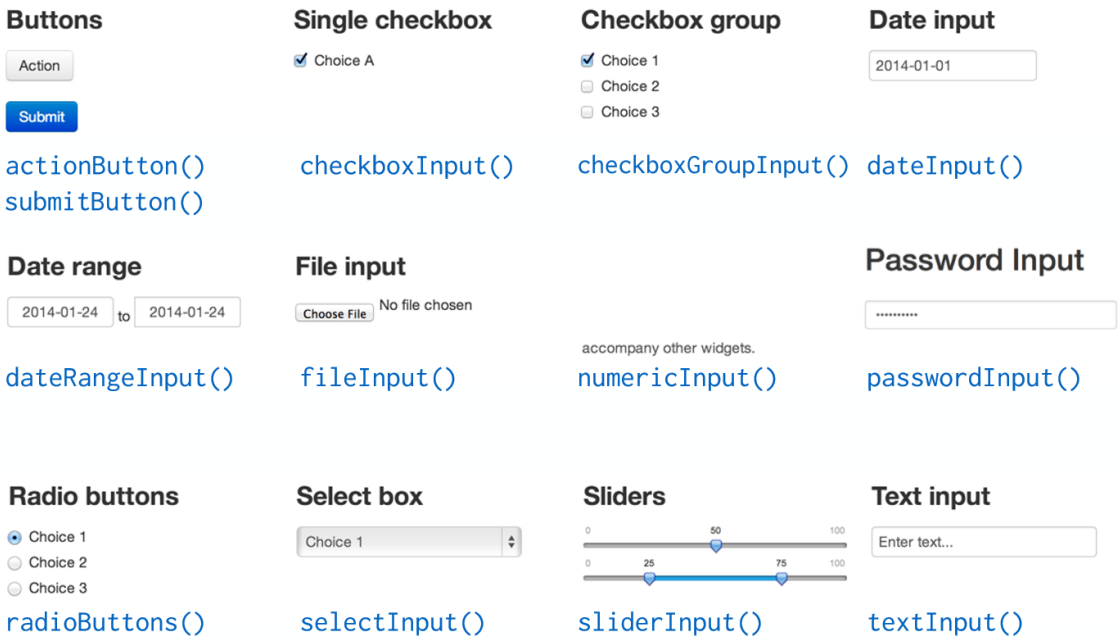
3.8 Sharing ui <-> server

Server and ui only communicates through inputs and outputs

- We can add an other file **global.R** if we want to share elements (datasets, functions. . .) between **UI** and **SERVER**
- All the elements in **global.R** are available for **ui.R** and **server.R**
- The script **global.R** is running just one time, at the beginning of the process.

4 Inputs

4.1 Global view



4.2 Numeric

- Function:

```
numericInput(inputId, label, value, min = NA, max = NA, step = NA)
```

- Example:

```
numericInput(inputId = "id_num", label = "Please select a number",
         value = 0, min = 0, max = 100, step = 10)
```

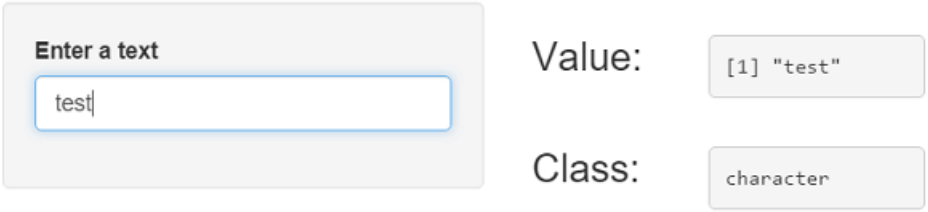
4.3 Characters

- Function:

```
textInput(inputId, label, value = "")
```

- Example:

```
textInput(inputId = "id_txt", label = "Enter a text", value = "")
```



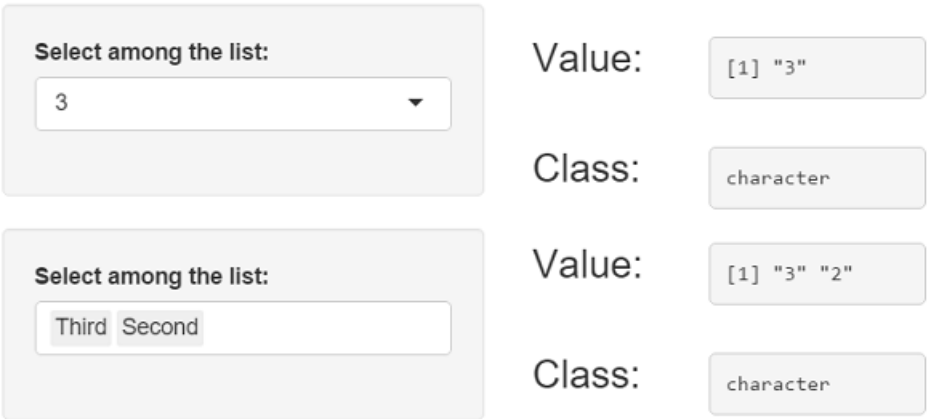
4.4 Unique choice in a list

- Function:

```
selectizeInput(inputId, label, choices, selected = NULL, multiple = FALSE,
        selectize = TRUE, width = NULL, size = NULL)
```

- Example:

```
selectizeInput(inputId = "id_sel1", label = "Select among the list: ", selected = 3,
           choices = c(1:3))
```



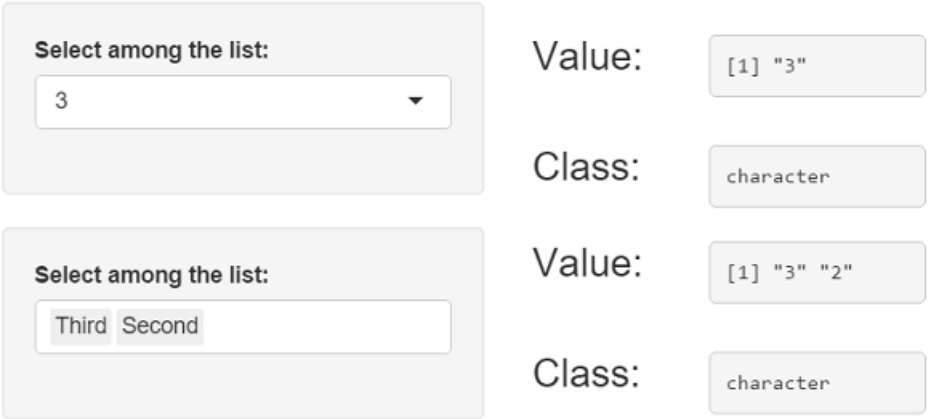
4.5 Multiple choices in a list

- Function:

```
selectInput(inputId, label, choices, selected = NULL, multiple = FALSE,
        selectize = TRUE, width = NULL, size = NULL)
```

- Example:

```
selectInput(inputId = "id_sel2", label = "Select among the list: ", selected = 3,
        choices = c("First" = 1, "Second" = 2, "Third" = 3), multiple = TRUE)
```



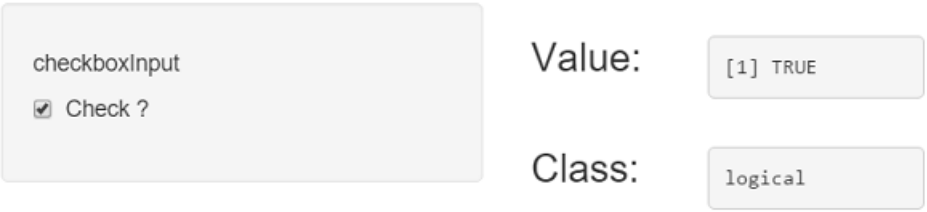
4.6 Simple Checkbox

- Function:

```
checkboxInput(inputId, label, value = FALSE)
```

- Example:

```
checkboxInput(inputId = "id_check_1", label = "Check?")
```



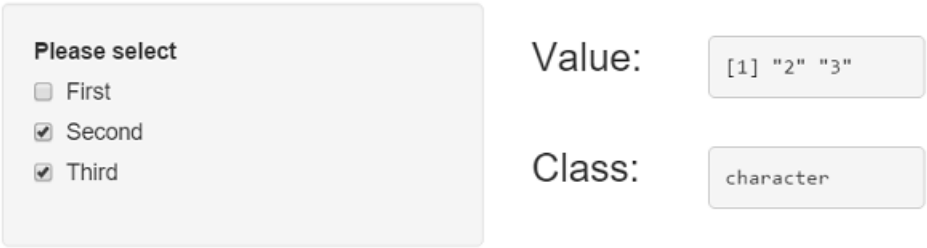
4.7 Multiple checkboxes

- Function:

```
checkboxGroupInput(inputId, label, choices, selected = NULL, inline = FALSE)
```

- Example:

```
checkboxGroupInput(inputId = "id_check_2", label = "Please select", selected = 3,
                   choices = c("First" = 1, "Second" = 2, "Third" = 3))
```



4.8 Radio buttons

- Function:

```
radioButtons(inputId, label, choices, selected = NULL, inline = FALSE)
```

- Example:

```
radioButtons(inputId = "id_radio", label = "Select one",
         choices = c("First" = 1, "Second" = 2, "Third" = 3),
         selected = 3)
```

4.9 Date | *Code*

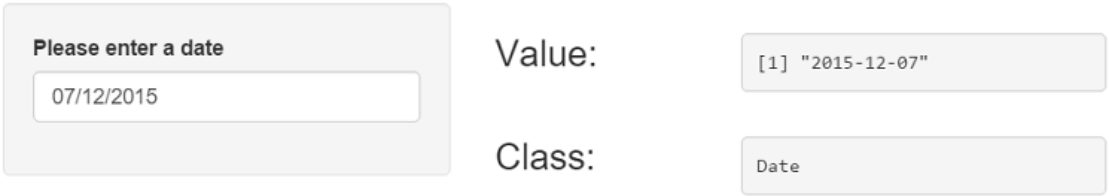
- Function:

```
dateInput(inputId, label, value = NULL, min = NULL, max = NULL, format = "yyyy-mm-dd",
      startview = "month", weekstart = 0, language = "en")
```

- Example:

```
dateInput(inputId = "id_date", label = "Please enter a date", value = "09/10/2020",
      format = "dd/mm/yyyy", startview = "month", weekstart = 0, language = "fr")
```

4.10 Date | *App*



4.11 Period | *Code*

- Function:

```
dateRangeInput(inputId, label, start = NULL, end = NULL, min = NULL, max = NULL,
           format = "yyyy-mm-dd", startview = "month", weekstart = 0,
           language = "en", separator = " to ")
```

- Example:

```
dateRangeInput(inputId = "id_daterange", label = "Please Select a date range",
           start = "2020-10-04", end = "2020-10-18", format = "yyyy-mm-dd",
           language = "en", separator = " to ")
```

4.12 Period | *App*

4.13 Numeric slider numérique: one value

- Function:

```
sliderInput(inputId, label, min, max, value, step = NULL, round = FALSE,
        format = NULL, locale = NULL, ticks = TRUE, animate = FALSE,
        width = NULL, sep = ",", pre = NULL, post = NULL)
```

- Example:

```
sliderInput(inputId = "id_slider", label = "Select a number", min = 0, max = 10,
        value = 5, step = 1)
```

4.14 Numeric slider: range

- Function:

```
sliderInput(inputId, label, min, max, value, step = NULL, round = FALSE,
        format = NULL, locale = NULL, ticks = TRUE, animate = FALSE,
        width = NULL, sep = ",", pre = NULL, post = NULL)
```

- Example:

```
sliderInput(inputId = "id_slider2", label = "Select a number", min = 0, max = 10,
        value = c(2,7), step = 1)
```

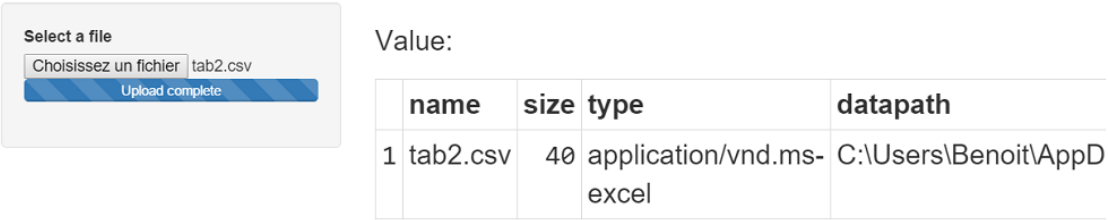
4.15 Importing a file

- Function:

```
fileInput(inputId, label, multiple = FALSE, accept = NULL)
```

- Example:

```
fileInput(inputId = "id_file", label = "Select a file")
```



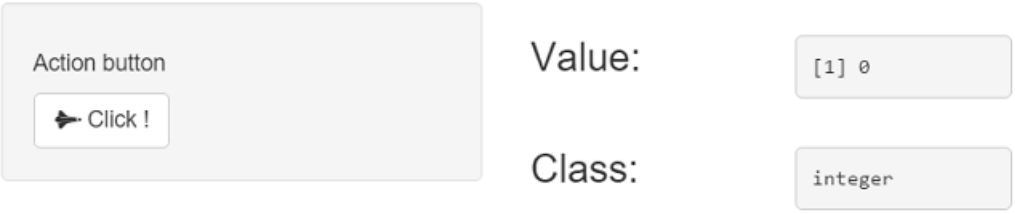
4.16 Action button

- Function:

```
actionButton(inputId, label, icon = NULL, ...)
```

- Example:

```
actionButton(inputId = "id_action", label = "Click !",
         icon = icon("hand-spock-o"))
```



4.17 Taking things further: building an input

Require skills in HTML/CSS/JavaScript

Tutorial: http://shiny.rstudio.com/articles/building-inputs.html

Two examples:

- http://shiny.rstudio.com/gallery/custom-input-control.html
- http://shiny.rstudio.com/gallery/custom-input-bindings.html

5 Outputs

5.1 Global view

server fonction	ui fonction	type de sortie
renderDataTable()	dataTableOutput()	une table intéractive
renderImage()	imageOutput()	une image sauvegardée
renderPlot()	plotOutput	un graphique R
renderPrint()	verbatimTextOutput()	affichage type console R
renderTable()	tableOutput()	une table statique
renderText()	textOutput()	une chaîne de caractère
renderUI()	uiOutput()	un élément de type UI

5.2 Rules to define outputs

- assign the output in the list **output**, use a good name to identify it in the **UI**
- use a function **renderXX({expr})**

```
#ui.R
selectInput("lettre", "Lettres:", LETTERS[1:3])
verbatimTextOutput(outputId = "selection")
#server.R
output$selection <- renderPrint({input$lettre})
```

5.3 Print

- **ui.r**:

```
verbatimTextOutput(outputId = "texte")
```

- **server.r**:

```
output$texte <- renderPrint({
  c("Hello shiny !")
})
```

```
[1] "Hello shiny !"
```

5.4 Text

- **ui.r**:

```
textOutput(outputId = "texte")
```

- **server.r**:

```
output$texte <- renderText({
  c("Hello shiny !")
})
```

Hello shiny !

5.5 Plot | *Code*

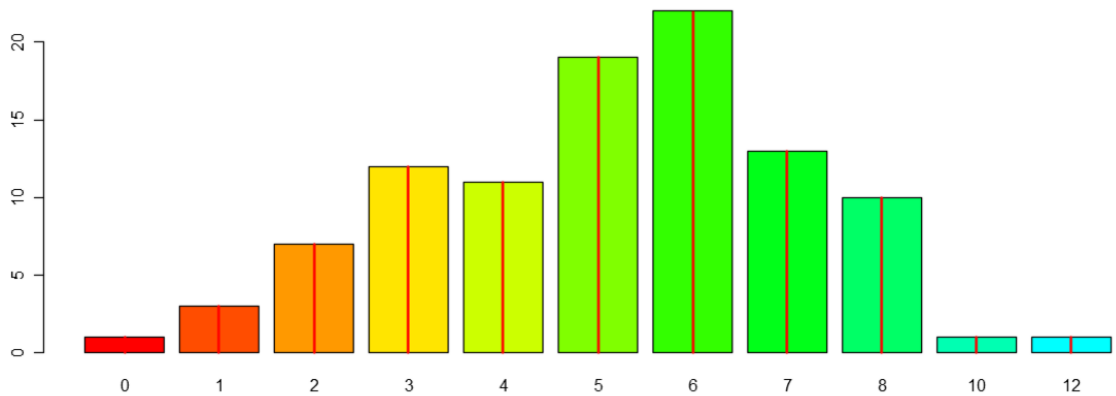
- **ui.r**:

```
plotOutput("myplot")
```

- **server.r**:

```
output$myplot <- renderPlot({
  require(grDevices) # for colours
  tN <- table(Ni <- stats::rpois(100, lambda = 5))

  r <- barplot(tN, col = rainbow(20))
  lines(r, tN, type = "h", col = "red", lwd = 2)
})
```

5.6 Plot | *App*



5.7 Table | *Code*

- **ui.r**:

```
tableOutput(outputId = "table")
```

- **server.r**:

```
data("iris")
output$table <- renderTable({
  iris[1:5, ]
})
```

5.8 Table | *App*

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.10	3.50	1.40	0.20	setosa
2	4.90	3.00	1.40	0.20	setosa
3	4.70	3.20	1.30	0.20	setosa
4	4.60	3.10	1.50	0.20	setosa
5	5.00	3.60	1.40	0.20	setosa

5.9 DataTable | *Code*

- **ui.r**:

```
dataTableOutput(outputId = "dataTable")
```

- **server.r**:

```
data("iris")
output$dataTable <- renderDataTable({
  iris
})
```

5.10 DataTable | *App*

Show 25 entries Search:

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
5.1	3.5	1.4	0.2	setosa
4.9	3.0	1.4	0.2	setosa
4.7	3.2	1.3	0.2	setosa
4.6	3.1	1.5	0.2	setosa
5.0	3.6	1.4	0.2	setosa
Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species

Showing 1 to 5 of 5 entries Previous 1 Next

5.11 Defining UI elements in the SEVER | Process

- **Sometimes it could be interested to define inputs in the server**
- For instance when an input **depends on another input**
- We can do that with `uiOutput` and `renderUI`

5.12 Defining UI elements in the SEVER | A simple example

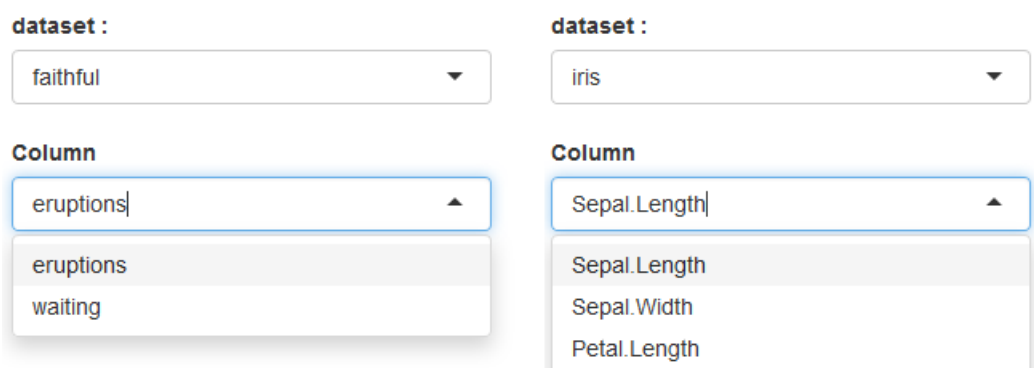
- **ui.r**:

```
uiOutput(outputId = "columns")
```

- **server.r**:

```
output$columns <- renderUI({
  selectInput(inputId = "sel_col", label = "Column", choices = colnames(data))
})
```

```
shinyApp(
  ui = fluidPage(
    selectInput(inputId = "my_data", label = "dataset: ",
                choices = c("iris", "faithful")),
    uiOutput(outputId = "columns")
  ),
  server = function(input, output) {
    data <- reactive(get(input$my_data, "package:datasets"))
    output$columns <- renderUI({
      selectInput(inputId = "sel_col", label = "Column", choices = colnames(data()))
    })
  })
```



5.13 Taking things further: building an output

Require some skills in HTML/CSS/JavaScript

Tutorial: http://shiny.rstudio.com/articles/building-outputs.html

6 Organizing the application

6.1 sidebarLayout | Definition

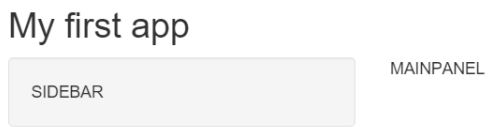
Basic template `sidebarLayout` divides the page in two columns and should contains:

- `sidebarPanel`, left part, generally for the inputs
- `mainPanel`, right part, generally for the outputs

```
shinyUI(
  fluidPage(
    titlePanel("Old Faithful Geyser Data"), # title
    sidebarLayout(
      sidebarPanel("Elements of sidebar (separated with commas)"),
```

```
      mainPanel("Elements of panel (separated with commas)")
    )
  )
)
```

6.2 sidebarLayout | Example

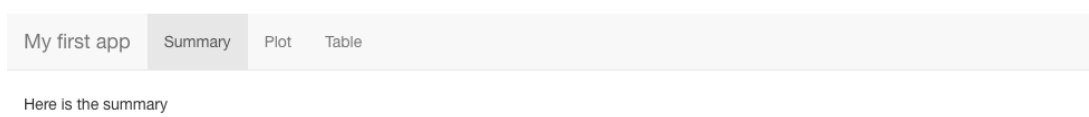


6.3 navbarPage | Definition

Use a navigation bar page with `navbarPage` and `tabPanel`:

```
shinyUI(
  navbarPage(
    title = "My first app",
    tabPanel(title = "Summary",
             "Here is the summary"),
    tabPanel(title = "Plot",
             "some charts"),
    tabPanel(title = "Table",
             "some tables")
  )
)
```

6.4 navbarPage | App



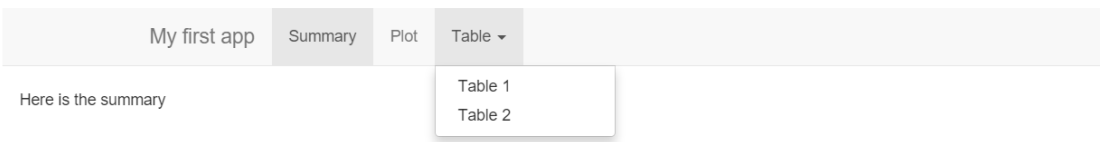
6.5 navbarPage | with `navbarMenu`

We can add a second level for the navigation with `navbarMenu`:

```
shinyUI(
  navbarPage(
```

```
    title = "My first app",
    tabPanel(title = "Summary",
             "Here is the summary"),
    tabPanel(title = "Plot",
             "some charts"),
    navbarMenu("Table",
      tabPanel("a table"),
      tabPanel("another table")
    )
  )
)
```

6.6 navbarPage | Shiny app



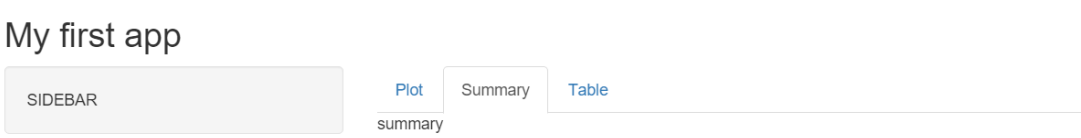
6.7 tabsetPanel | Definition

More generally, we can create navigation bar pages everywhere with `tabsetPanel` & `tabPanel`:

```
sidebarLayout(
  sidebarPanel("SIDEBAR"),
  mainPanel(
    tabsetPanel(
      tabPanel("Plot", plotOutput("plot")),
      tabPanel("Summary", verbatimTextOutput("summary")),
      tabPanel("Table", tableOutput("table"))
    )
  )
)
```

- **navbarPage**: create tabs in the application
- **tabsetPanel**: create tabs in a structure of the application

6.8 tabsetPanel | Example

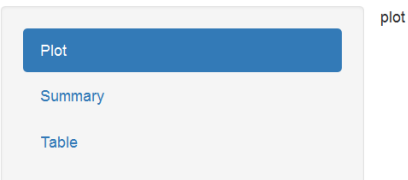


6.9 navlistPanel | Definition

An alternative to `tabsetPanel`, to obtain a vertical position instead of horizontal: `navlistPanel`

```
shinyUI(fluidPage(
      navlistPanel(
        tabPanel("Plot", plotOutput("plot")),
        tabPanel("Summary", verbatimTextOutput("summary")),
        tabPanel("Table", tableOutput("table"))
    )
))
```

6.10 navlistPanel | Example



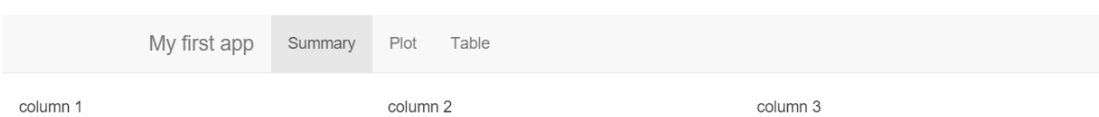
6.11 Grid Layout | Definition

Define your own organization with `fluidRow()` and `column()`

- any lines can be divided into 12 columns
- page size fits automatically to the number of rows/columns.

```
tabPanel(title = "Summary",
     # A fluid row can contain from 0 to 12 columns
     fluidRow(
       # A column is defined necessarily
       # with its argument "width"
       column(width = 4, "column 1"),
       column(width = 4, "column 2"),
       column(width = 4, "column 3"),
     ))
```

6.12 Grid Layout | Shiny app



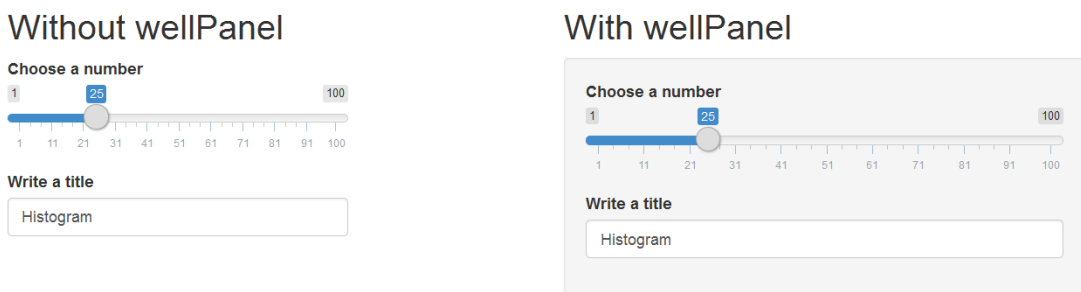
6.13 wellPanel | Definition

You can obtain a grey background with `wellPanel`:

```
fluidRow(
  column(6,
```

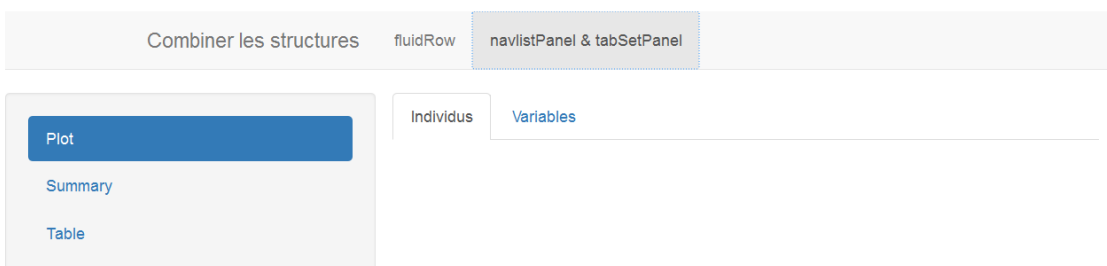
```
    h2("Without wellPanel"), # title
    sliderInput("num", "Choose a number", value = 25, min = 1, max = 100),
    textInput("title", value = "Histogram", label = "Write a title")
  ),
  column(6,
    h2("With wellPanel"), # title
    wellPanel(
      sliderInput("num", "Choose a number", value = 25, min = 1, max = 100),
      textInput("title", value = "Histogram", label = "Write a title")
    )
  )
)
```

6.14 wellPanel | Example



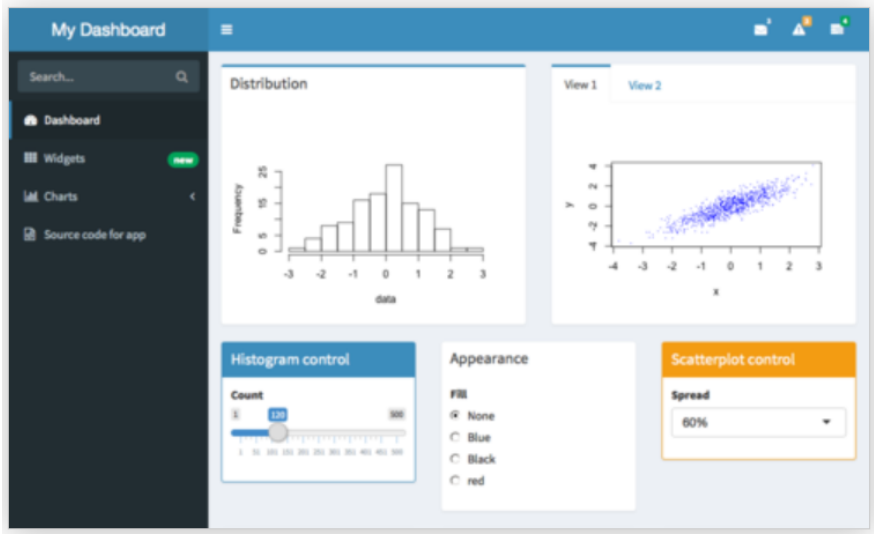
6.15 Combine structures | Shiny app

All structures can be used at the same time!



6.16 shinydashboard

The package shinydashboard has other functions to define `dashboards`:



https://rstudio.github.io/shinydashboard/

7 Interactive charts

7.1 Introduction

Since the creation of htmlwidgets package, more and more javascript possibilities are available with **R**:

- dygraphs (time series)
- DT (interactive tables)
- Leafet (maps)
- d3heatmap
- rAmCharts
- visNetwork
- ...

You can look at this gallery

7.2 Integration in shiny

All these packages can be used in **shiny**. Indeed, they posses the two required functions:

- **renderXX**
- **xxOutput**

An example with dygraphs package:

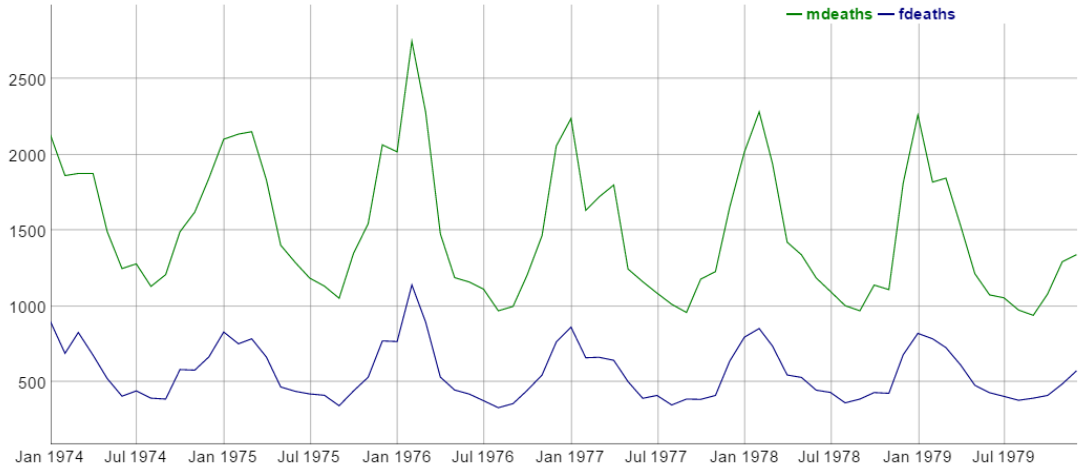
```
# Server
output$dygraph <- renderDygraph({
    dygraph(predicted(), main = "Predicted Deaths/Month")
})
# Ui
dygraphOutput("dygraph")
```

7.3 Examples for server and ui functions

Package	server function	ui function
dygraph	renderDygraph	dygraphOutput
rAmcharts	renderAmCharts	amChartsOutput
leaflet	renderLeaflet	leafletOutput
plotly	renderPlotly	plotlyOutput
visNetwork	renderVisNetwork	visNetworkOutput

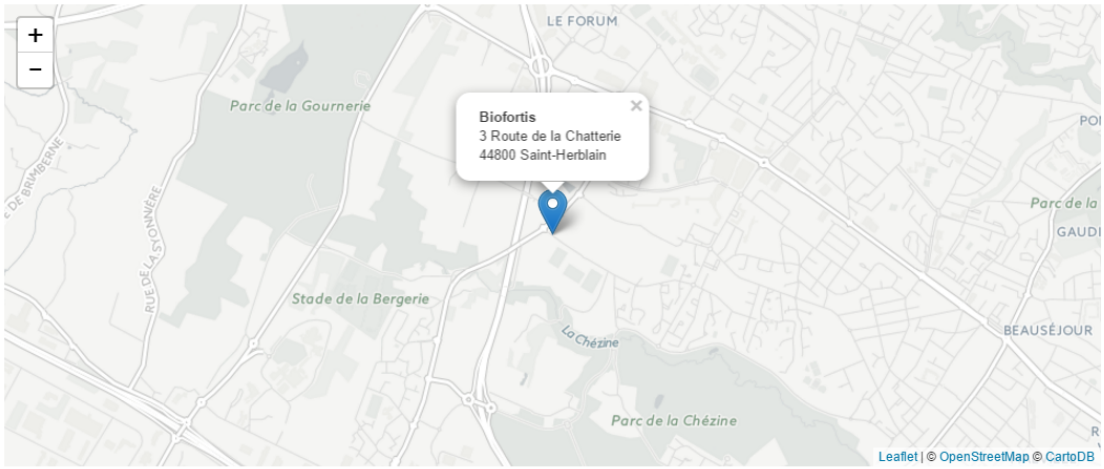
7.4 Interactive charts: example

dygraphs

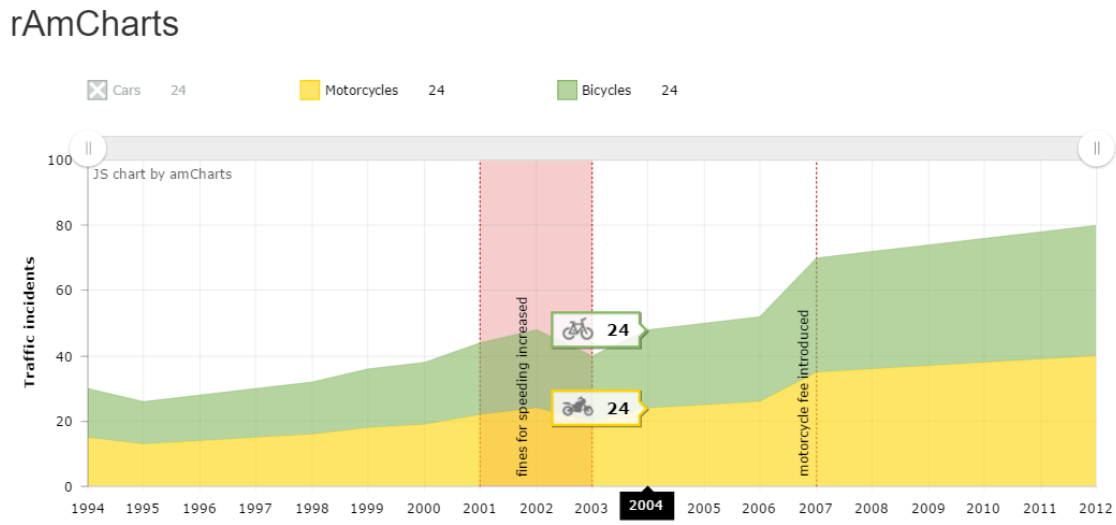


7.5 Interactive charts: example

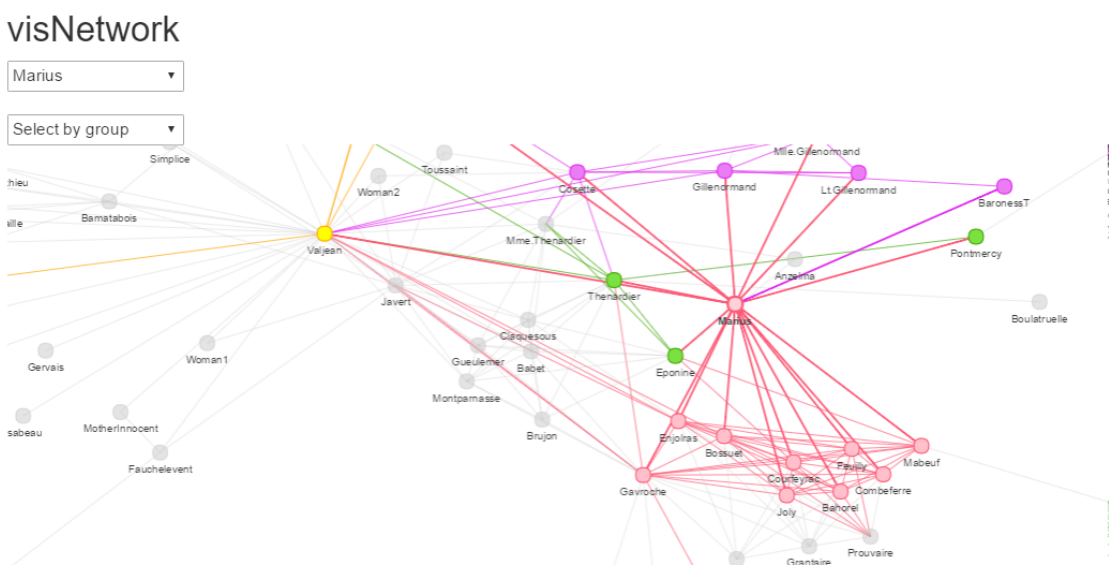
leaflet



7.6 Interactive charts: example



7.7 Interactive charts: example



8 Isolation

8.1 Definition

- By default, outputs and reactive expressions are updated as soon as the user changes one input
- It would be interesting to control this **update process**
- For instance, with a check button (**actionButton**) to start the computation of the outputs
- An input can be isolated with `isolate(input$id)`
- For an expression we use `isolate({expr})` (don't forget `{}`)

8.2 Example | ui.R

Three inputs: **color** and **bins** for the histogram, and one **actionButton**:

```r
shinyUI(fluidPage(
  titlePanel("Isolation"),
  sidebarLayout(
    sidebarPanel(
      radioButtons(inputId = "col", label = "Choose a color", inline = TRUE,
                   choices = c("red", "blue", "darkgrey")),
      sliderInput("bins", "Number of bins:", min = 1, max = 50, value = 30),
      actionButton("go_graph", "Update!")
    ),
    mainPanel(plotOutput("distPlot"))
  )
))
```

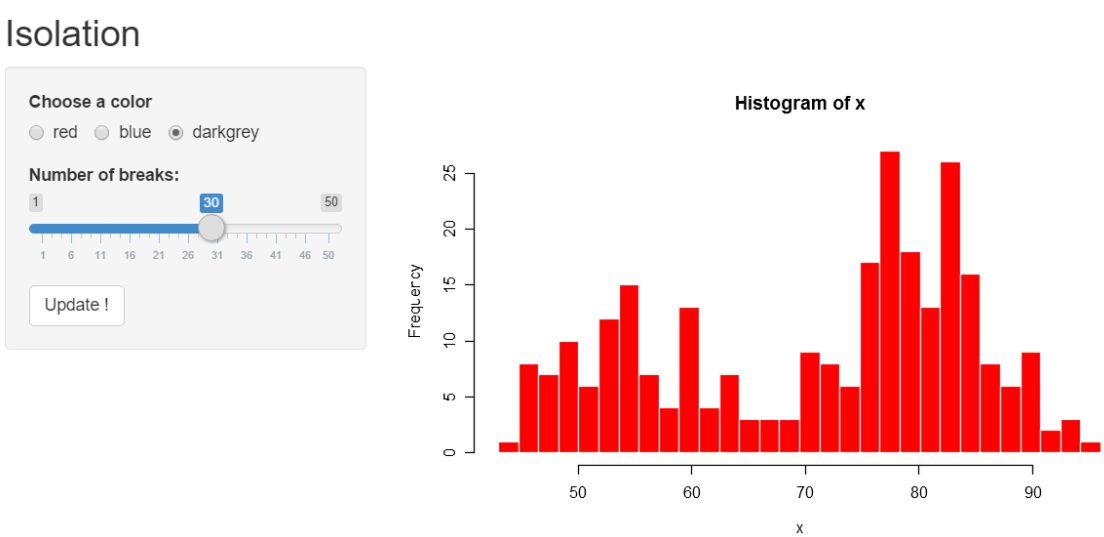
8.3 Example | server.R

Isolation of everything excepted the **actionButton**:

```r
shinyServer(function(input, output) {
  output$distPlot <- renderPlot({
    input$go_graph #action to start the update process
    isolate({
      inputColor <- input$color
      x <- faithful[, 2]
      bins <- seq(min(x), max(x), length.out = input$bins + 1)
      hist(x, breaks = bins, col = inputColor, border = 'white')
    })
  })
})
```

The histogram will be updated when the user will click on the button.

8.4 Example | App



9 Reactive expressions

9.1 Definition

- Very usefull when we want to use the same result/objects in many outputs, by doing the calculation just one time.
- Just have to use the function `reactive` in **server.R**
- For instance, we want to visualize two graphs of a PCA:
  - projection of individuals
  - projection of variables.

9.2 Without reactive expressions

- **server.R**: the calculation is performed twice...
- By default, **only** R expressions in **renderXX** functions are updated.

```
require(FactoMineR) ; data("decathlon")

output$graph_pca_ind <- renderPlot({
  res_pca <- PCA(decathlon[ ,input$variables], graph = FALSE)
  plot.PCA(res_pca, choix = "ind", axes = c(1,2))
})

output$graph_pca_var <- renderPlot({
  res_pca <- PCA(decathlon[,input$variables], graph = FALSE)
  plot.PCA(res_pca, choix = "var", axes = c(1,2))
})
```

9.3 With a reactive expression

- **server.R** : The calculation is performed only once!

```
require(FactoMineR) ; data("decathlon")

res_pca <- reactive({
  PCA(decathlon[,input$variables], graph = FALSE)
})

output$graph_pca_ind <- renderPlot({
  plot.PCA(res_pca(), choix = "ind", axes = c(1,2))
})

output$graph_pca_var <- renderPlot({
  plot.PCA(res_pca(), choix = "var", axes = c(1,2))
})
```

9.4 Notes

- A reactive expression will save time and memory.
- **Use reactive expressions only when they depend on inputs**
- Reactive expressions updates as soon as the user changes an input
- We obtain its value with "()"

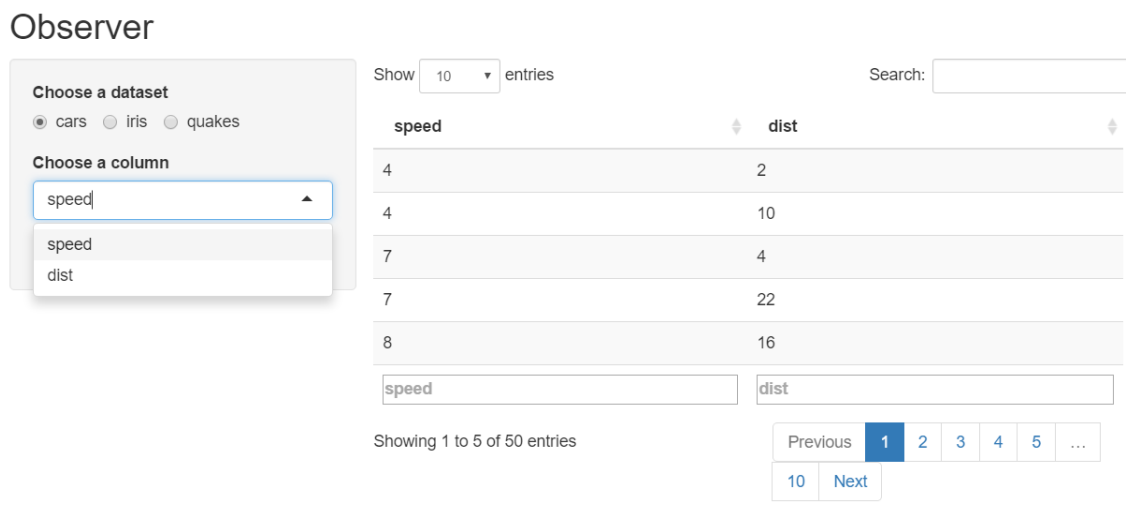
10 Observe & functions to update

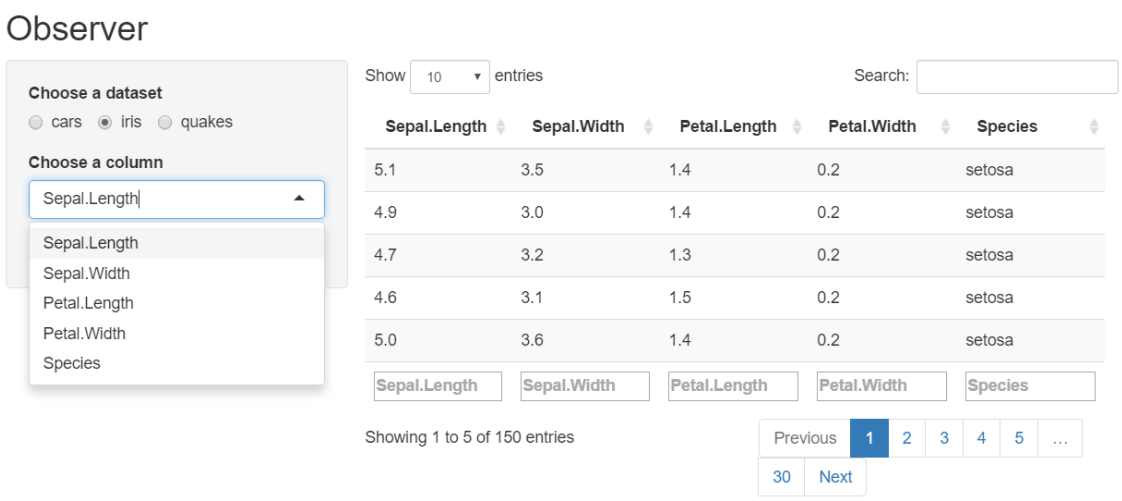
10.1 Observe & fonctions to update

- There exists many functions to update inputs and some structures
- They start with `update...`
- They are generally used in `observe({expr})`
- **Be careful**: we have to add *"session"* in the definition of **server**

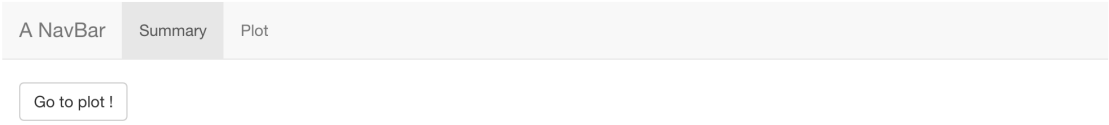
```
shinyServer(function(input, output, session) {...})
```

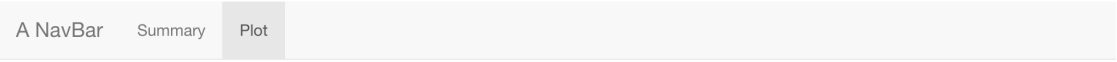
10.2 Example for an input | App

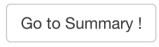




10.3 Example for tabs | App







For inputs:

- **updateCheckboxGroupInput**
- **updateCheckboxInput**
- **updateDateInput Change**
- **updateDateRangeInput**
- **updateNumericInput**
- **updateRadioButtons**
- **updateSelectInput**
- **updateSelectizeInput**
- **updateSliderInput**
- **updateTextInput**

To change a selected tab

- **updateNavbarPage**, **updateNavlistPanel**, **updateTabsetPanel**

10.4 Example for an input | ui.R

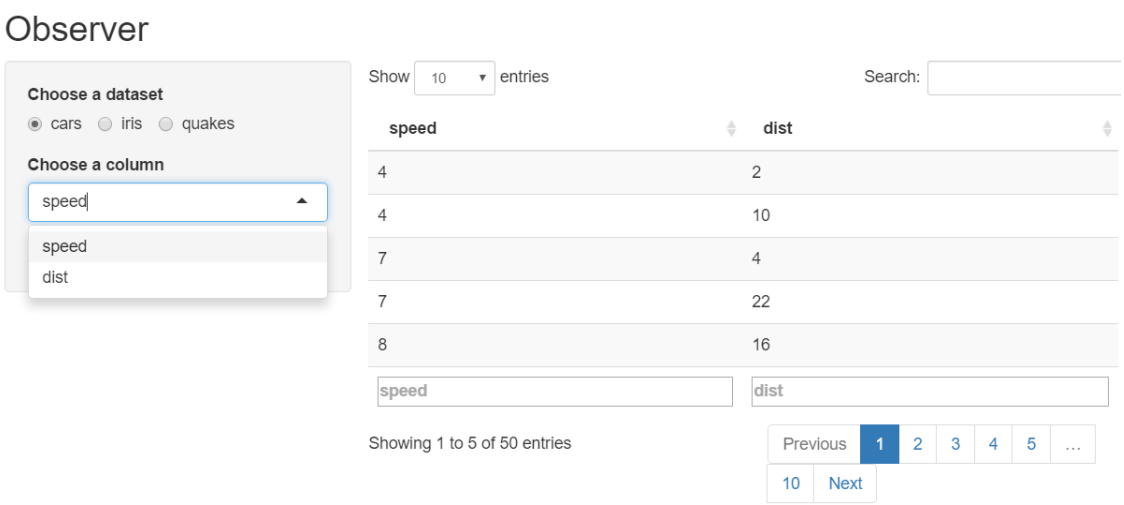
```
shinyUI(fluidPage(
  titlePanel("Observe"),
  sidebarLayout(
    sidebarPanel(
      radioButtons(inputId = "id_dataset", label = "Choose a dataset", inline = TRUE,
                   choices = c("cars", "iris", "quakes"), selected = "cars"),
      selectInput("id_col", "Choose a column", choices = colnames(cars)),
      textOutput(outputId = "txt_obs")
    ),
    mainPanel(fluidRow(
      dataTableOutput(outputId = "dataset_obs")
    ))
```

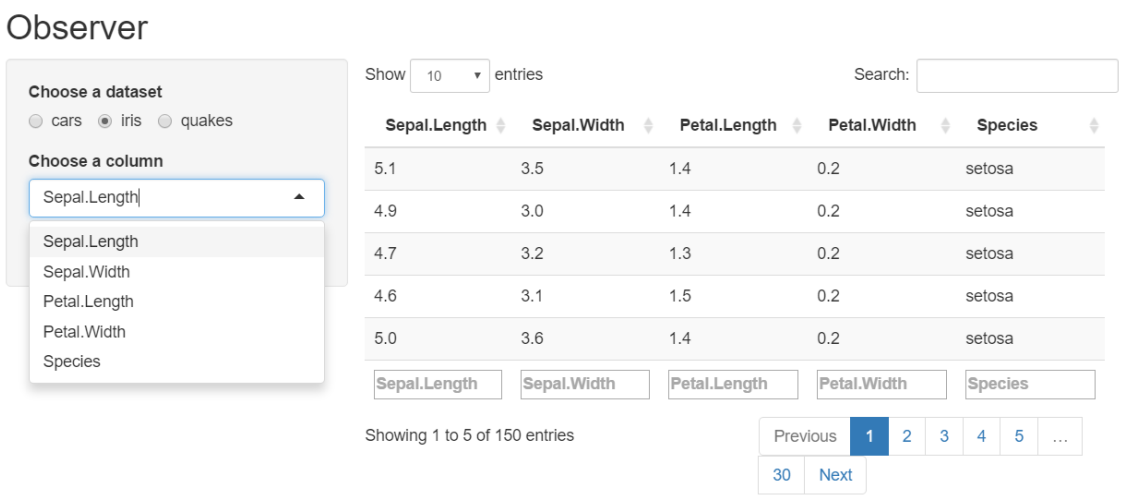
```
  )
))
```

10.5 Example for an input | server.R

```
shinyServer(function(input, output, session) {
  dataset <- reactive(get(input$id_dataset, "package:datasets"))

  observe({
    updateSelectInput(session, inputId = "id_col", label = "Choose a column",
                      choices = colnames(dataset()))
  })

  output$txt_obs <- renderText(paste0("Selected column : ", input$id_col))

  output$dataset_obs <- renderDataTable(
    dataset(),
    options = list(pageLength = 5)
  )
})
```

10.6 Example for an input | App





10.7 Example for tabs | ui.R

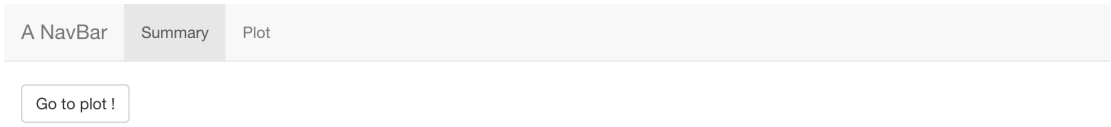
We have to add an ID in the structure

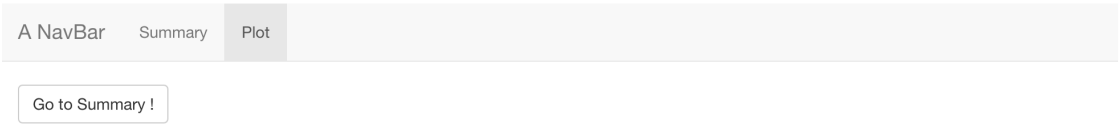
```
shinyUI(
  navbarPage(
    id = "idnavbar", # need an id for observe & update
    title = "A NavBar",
    tabPanel(title = "Summary",
             actionButton("goPlot", "Go to plot !")),
    tabPanel(title = "Plot",
             actionButton("goSummary", "Go to Summary !"))

  )
)
```

10.8 Example for tabs | server.R

```
shinyServer(function(input, output, session) {
  observe({
    input$goPlot #action to start the update process
    updateTabsetPanel(session, "idnavbar", selected = "Plot")
  })
  observe({
    input$goSummary #action to start the update process
    updateTabsetPanel(session, "idnavbar", selected = "Summary")
  })
})
```

10.9 Example for tabs | App





10.10 ObserveEvent

- An alternative to `observe`: `observeEvent`
- We have to define both the expression of the event and the expression to execute when the event occurs

```
# with observe
observe({
  input$goPlot
  updateTabsetPanel(session, "idnavbar", selected = "Plot")
})

# same with observeEvent
observeEvent(input$goSummary, {
  updateTabsetPanel(session, "idnavbar", selected = "Summary")
})
```

11 Conditional panels

11.1 Definition

- We can use conditions to print some inputs/outputs

`conditionalPanel(condition = [...], )`

- The condition can depend on inputs or outputs
- Be careful: it should be written in **javascript**...

`conditionalPanel(condition = "input.checkbox == true", [...])`

11.2 Example for an input

```
shinyApp(
  ui = fluidPage(
    fluidRow(
      column(width = 4, align = "center",
        checkboxInput("checkbox", "View other inputs", value = FALSE)
      ),
      column(width = 8, align = "center",
        conditionalPanel(
          condition = "input.checkbox == true",
          sliderInput("slider", "Select value", min = 1, max = 10, value = 5),
          textInput("txt", "Enter text", value = "")
        )
      )
    )
  ),
  server = function(input, output) {}
)
```

11.3 Example for an input

12 Taking things further: HTML / CSS

12.1 Including HTML

Many **html tags** are availabe with `tags` functions:

```
##   [1] "a"                  "abbr"                "address"
##   [4] "animate"            "animateMotion"       "animateTransform"
##   [7] "area"               "article"             "aside"
##  [10] "audio"              "b"                   "base"
##  [13] "bdi"                "bdo"                 "blockquote"
##  [16] "body"               "br"                  "button"
##  [19] "canvas"             "caption"             "circle"
##  [22] "cite"               "clipPath"            "code"
##  [25] "col"                "colgroup"            "color-profile"
##  [28] "command"            "data"                "datalist"
##  [31] "dd"                 "defs"                "del"
```

```
##  [34] "desc"                "details"             "dfn"
##  [37] "dialog"              "discard"             "div"
##  [40] "dl"                  "dt"                  "ellipse"
##  [43] "em"                  "embed"               "eventsource"
##  [46] "feBlend"             "feColorMatrix"       "feComponentTransfer"
##  [49] "feComposite"         "feConvolveMatrix"    "feDiffuseLighting"
##  [52] "feDisplacementMap"   "feDistantLight"      "feDropShadow"
##  [55] "feFlood"             "feFuncA"             "feFuncB"
##  [58] "feFuncG"             "feFuncR"             "feGaussianBlur"
##  [61] "feImage"             "feMerge"             "feMergeNode"
##  [64] "feMorphology"        "feOffset"            "fePointLight"
##  [67] "feSpecularLighting"  "feSpotLight"         "feTile"
##  [70] "feTurbulence"        "fieldset"            "figcaption"
##  [73] "figure"              "filter"              "footer"
##  [76] "foreignObject"       "form"                "g"
##  [79] "h1"                  "h2"                  "h3"
##  [82] "h4"                  "h5"                  "h6"
##  [85] "hatch"               "hatchpath"           "head"
##  [88] "header"              "hgroup"              "hr"
##  [91] "html"                "i"                   "iframe"
##  [94] "image"               "img"                 "input"
##  [97] "ins"                 "kbd"                 "keygen"
## [100] "label"               "legend"              "li"
## [103] "line"                "linearGradient"      "link"
## [106] "main"                "map"                 "mark"
## [109] "marker"              "mask"                "menu"
## [112] "meta"                "metadata"            "meter"
## [115] "mpath"               "nav"                 "noscript"
## [118] "object"              "ol"                  "optgroup"
## [121] "option"              "output"              "p"
## [124] "param"               "path"                "pattern"
## [127] "picture"             "polygon"             "polyline"
## [130] "pre"                 "progress"            "q"
## [133] "radialGradient"      "rb"                  "rect"
## [136] "rp"                  "rt"                  "rtc"
## [139] "ruby"                "s"                   "samp"
## [142] "script"              "section"             "select"
## [145] "set"                 "slot"                "small"
## [148] "solidcolor"          "source"              "span"
## [151] "stop"                "strong"              "style"
## [154] "sub"                 "summary"             "sup"
## [157] "svg"                 "switch"              "symbol"
## [160] "table"               "tbody"               "td"
## [163] "template"            "text"                "textarea"
## [166] "textPath"            "tfoot"               "th"
## [169] "thead"               "time"                "title"
## [172] "tr"                  "track"               "tspan"
## [175] "u"                   "ul"                  "use"
## [178] "var"                 "video"               "view"
## [181] "wbr"
```

We can also use **html** code with **HTML** function:

```
fluidPage(
  HTML("<h1>My Shiny App</h1>")
)
```

12.2 Some interested tags

- `div(..., align = "center")`: center elements
- `br()`: line break
- `hr()`: horizontal line
- `img(src="img/logo.jpg", title="Popup", width = "80%")`: insert an image in **www/img**
- `a(href="https://r2018-rennes.sciencesconf.org/", target="_blank", "Rencontres R")`: link to a website
- `a(href = './doc/guide.pdf', target="_blank", class = "btn", icon("download"), 'Télécharger le guide utilisateur')`: link to download a document in **www/doc**

12.3 CSS: introduction

Shiny use Bootstrap for the **CSS** part.

As for classical web development, we can change the **CSS** in three ways:

- link to a **.css file** in the directory `www`
- adding **CSS** in the **HTML** header
- using **CSS** codes in an element.

Priority order : 1. **CSS** codes in an element 2. **CSS** in the **HTML** header 3. **.css file**

We can also use the shinythemes package.

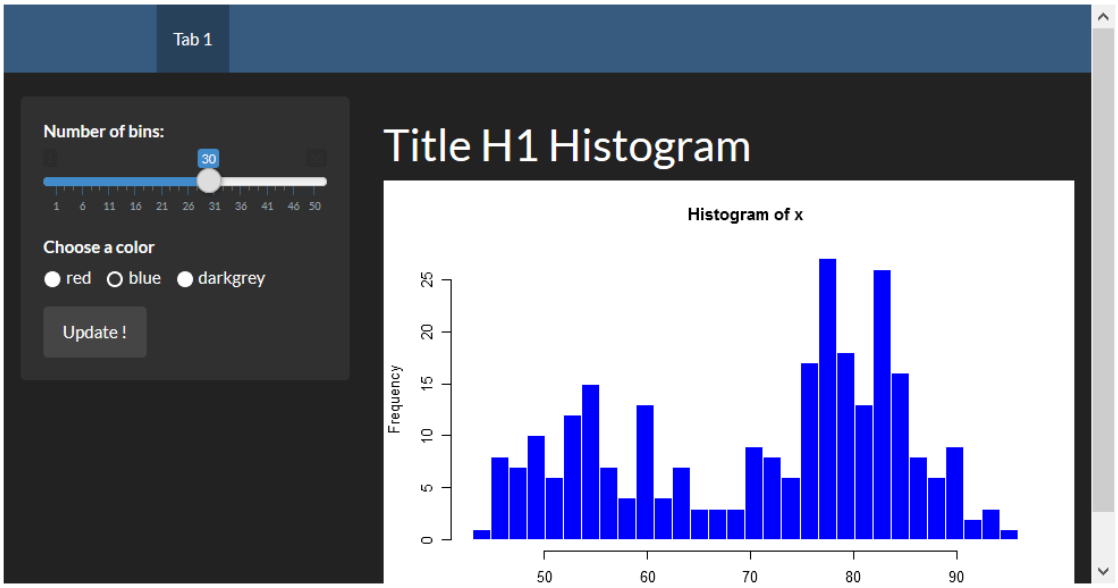
12.4 HTML / CSS | css external file

You can find some themes in bootswatch.

- Two ways to specify the theme: + option `theme` in some functions (`fluidPage`, `navbarPage`, ...) + with a **html tags** : `tags$head` et `tags$link`

```
library(shiny)
ui <- fluidPage(theme = "mytheme.css",
  # or with a tags
  tags$head(
tags$link(rel = "stylesheet", type = "text/css", href = "mytheme.css")
  ),
```

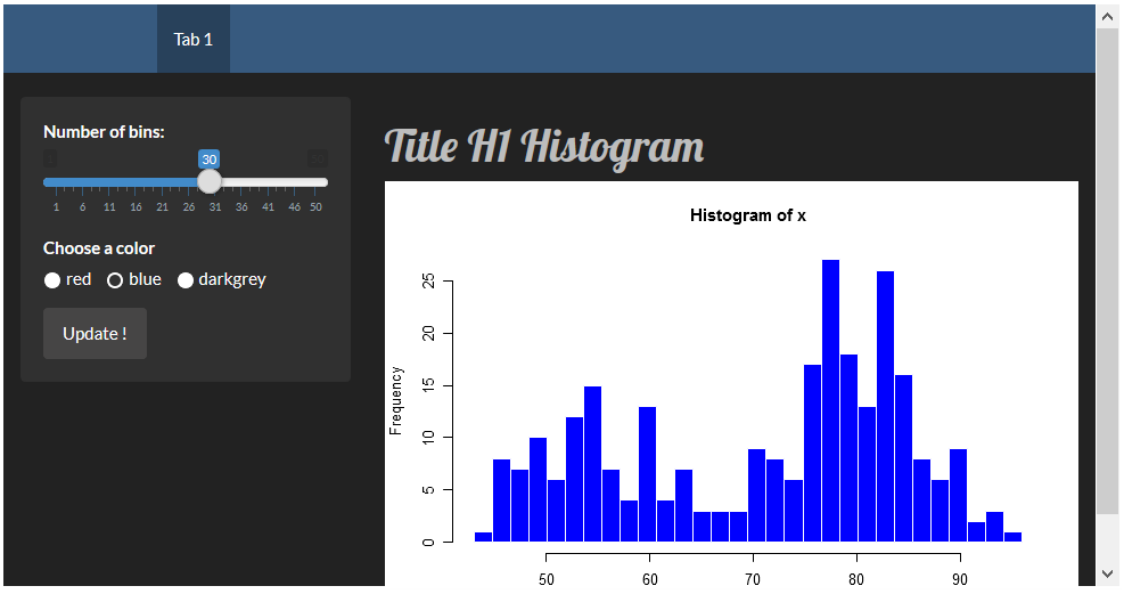
```
  # ...
)
```



12.5 HTML / CSS | css in the header

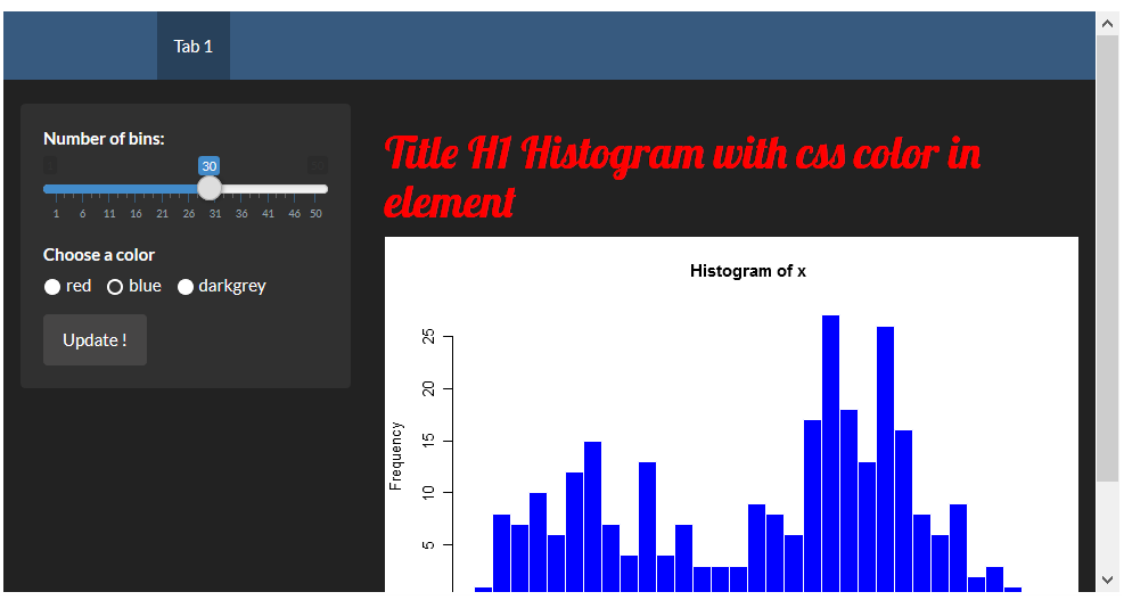
- With **html tags**: `tags$head` and `tags$style`

```
library(shiny)
  tags$head(
tags$style(HTML("h1 { color: #48ca3b;}")
)
  ),
  # ...
)
```



12.6 HTML / CSS | CSS in an element

```
library(shiny)
  h1("Mon titre", style = "color: #48ca3b;")
  # reste de l'application
)
```



13 Taking things further: some important "rules"

13.1 Good approach

- Choose underscore (_) instead of point (.) in the names of the objects or variables. Indeedt, the point . can lead to some confusions with other languages, such as **JavaScript**

- Use packrat package to avoid problems with **version packages**
- Use **R script** for the calculation part and make test with (testthat).

13.2 Good approach

- Divide the **ui.R** and **server.R** parts in several scripts, on for each tabs for instance:

```
# ui.R
shinyUI(
  navbarPage("Divide UI & SERVER",
source("src/ui/01_ui_plot.R", local = TRUE)$value,
source("src/ui/02_ui_data.R", local = TRUE)$value
  )
)
# server.R
shinyServer(function(input, output, session) {
  source("src/server/01_server_plot.R", local = TRUE)
  source("src/server/02_server_data.R", local = TRUE)
})
```

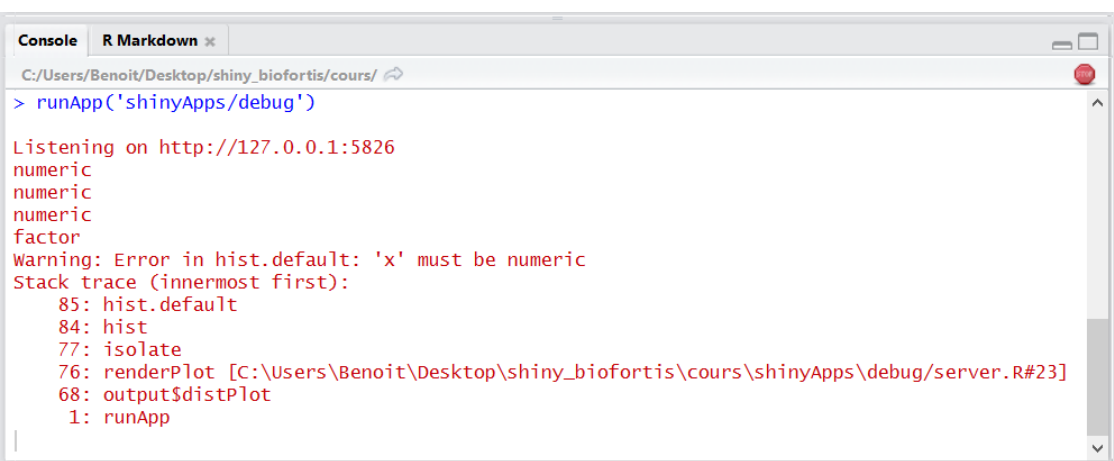
14 Taking things further: debugging

14.1 Printing in the console

- You can use some `print` in the application
- It allows to visualize informations during the process
- In **shiny**, use `cat(file=stderr(), ...)` to be sure that the display operates for all kind of outputs

```
 output$distPlot <- renderPlot({
  x <- iris[, input$variable]
  cat(file=stderr(), class(x)) # affichage de la classe de x
  hist(x)
})
```

14.2 Printing in the console



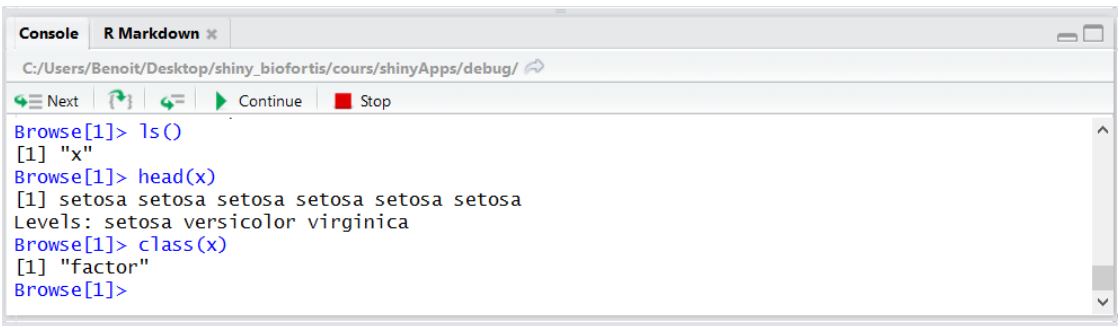
14.3 Manual launch of a browser

- We can launch a browser with `browser()` everywhere
- It allows to observe the different objects

```
 output$distPlot <- renderPlot({
  x <- iris[, input$variable]
  browser() # lancement du browser
  hist(x)
})
```

- Don't forget to remove it!

14.4 Manual launch of a browser



14.5 Automatic launch of a browser

- The option `options(shiny.error = browser)` allows to launch `browser()` as soon as an error appears

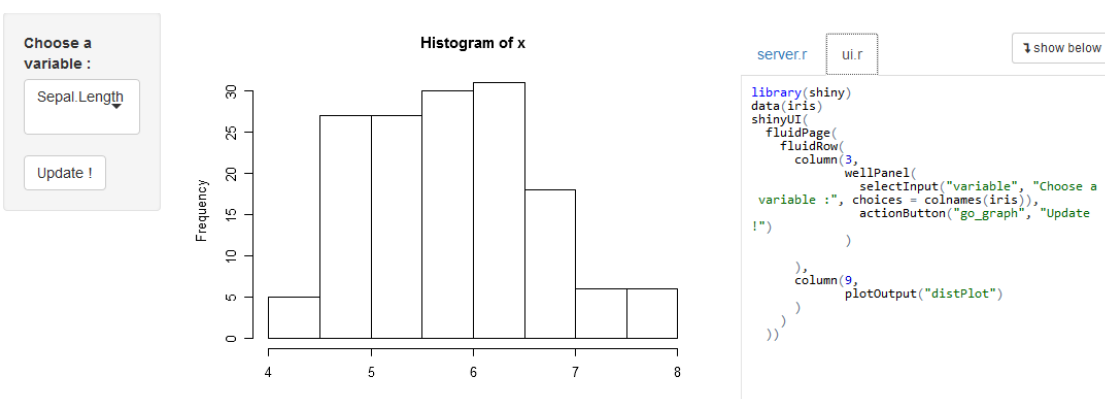
```
options(shiny.error = browser)
```

14.6 Mode "showcase"

- With the `display.mode="showcase"` in `runApp()`, we can observe directly the executed code:

```
runApp("path/to/myapp", display.mode="showcase")
```

14.7 Mode "showcase"



14.8 Reactive log

- With `shiny.reactlog`, we can visualize dependences between **reactive objects** and **shiny**
  - use `ctrl+F3` in the web browser
  - with `showReactLog()` in the shiny code

```
options(shiny.reactlog=TRUE)

output$distPlot <- renderPlot({
  x <- iris[, input$variable]
  showReactLog() # launch shiny.reactlog
  hist(x)
})
```

14.9 Reactive log

14.10 Communication in the server

- We can visualize these communications with the option `shiny.trace`

```
options(shiny.trace = TRUE)
```

14.11 Communication in the server

```
Console   R Markdown
C:/Users/Benoit/Desktop/shiny_biofortis/cours/
> runApp('shinyApps/debug')

Listening on http://127.0.0.1:5826
SEND {"config":{"workerId":"","sessionId":"d881eec9a56887dd66d5d6bf2f8776ed"}}
RECV {"method":"init","data":{"go_graph:shiny.action":0,"variable":"Sepal.Length",".clientdata_
output_distPlot_width":816,".clientdata_output_distPlot_height":400,".clientdata_output_distPlo
t_hidden":false,".clientdata_pixelratio":1,".clientdata_url_protocol":"http:",".clientdata_url_
hostname":"127.0.0.1",".clientdata_url_port":"5826",".clientdata_url_pathname":"/",".clientdata
_url_search":"",".clientdata_url_hash_initial":"",".clientdata_singletons":"",".clientdata_allo
wDataUriScheme":true}}
SEND {"custom":{"busy":"busy"}}
SEND {"custom":{"recalculating":{"name":"distPlot","status":"recalculating"}}}
SEND {"custom":{"recalculating":{"name":"distPlot","status":"recalculated"}}}
SEND {"custom":{"busy":"idle"}}
SEND {"errors":[],"values":{"distPlot":{"src":"data:image/png;[base64 data]","width":816,"heigh
t":400,"coordmap":[{"domain":{"left":3.84,"right":8.16,"bottom":-1.24,"top":32.24},"range":{"le
ft":59.04,"right":785.76,"bottom":325.56,"top":58.04},"log":{"x":null,"y":null},"mapping":{}}]}
},"inputMessages":[]}
RECV {"method":"update","data":{"variable":"Petal.Length"}}
```

14.12 Error tracking

- Sincce `shiny_0.13.1`, we can obtain a stack trace when an error occurs
- We can obtain more inofrmations with `options(shiny.fullstacktrace = TRUE)`

```
options(shiny.fullstacktrace = TRUE)
```

14.13 Error tracking

```
Console   R Markdown
C:/Users/Benoit/Desktop/shiny_biofortis/cours/
> runApp('shinyApps/debug')

Listening on http://127.0.0.1:5826
Warning: Error in hist.default: 'x' must be numeric
Stack trace (innermost first):
    88: h
    87: .handleSimpleError
    86: stop
    85: hist.default
    84: hist
    83: ..stacktraceon.. [C:\Users\Benoit\Desktop\shiny_biofortis\cours\shinyApps\debug/server.
R#35]
    82: contextFunc
    81: env$runWith
    80: withReactiveDomain
    79: ctx$run
```

15 References

15.1 Tutorials / Examples

- http://shiny.rstudio.com/
- http://shiny.rstudio.com/articles/
- http://shiny.rstudio.com/tutorial/
- http://shiny.rstudio.com/gallery/
- https://www.rstudio.com/products/shiny/shiny-user-showcase/

- http://www.showmeshiny.com/